

Stem-and-Leaf Display and Box-and-Whisker Plot

Worksheet # 1

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Stem-and-Leaf Display

1. The following data represents the processing times (in milliseconds) of a sorting algorithm:

12, 14, 18, 21, 24, 25, 28, 32, 33, 35, 38, 41, 43, 47, 50

1. Construct a **Stem-and-Leaf Display** for the given dataset.
 2. What is the **mode** of the dataset?
 3. What is the **range** of the dataset?
 4. How many data points are greater than 30?
 5. What is the **median** processing time?
2. A machine learning researcher records the execution times (in seconds) of an AI model on different datasets. The recorded times are:

2.1, 2.4, 2.6, 3.0, 3.3, 3.5, 3.8, 4.2, 4.4, 4.7, 5.1, 5.3, 5.8, 6.0, 6.3

1. Construct a **Stem-and-Leaf Display** for the given dataset.
2. What is the **mode** of the dataset?
3. What is the **range** of the dataset?
4. How many data points are greater than 4.0?
5. What is the **median** execution time?

Conceptual Questions

1. What advantage does a **Stem-and-Leaf Display** have over a frequency table?
2. How would you modify the Stem-and-Leaf Display if the dataset had values up to 12.3 seconds?
3. Can a Stem-and-Leaf Display help identify outliers? Explain.

Box-and-Whisker Plot

1. The response times (in milliseconds) for a web server under load are:

22, 25, 29, 30, 31, 35, 40, 42, 45, 47, 50, 52, 55, 60, 65

1. Find the **five-number summary** (Minimum, Q1, Median, Q3, Maximum).
2. Construct a **Box-and-Whisker Plot** for the given data.
3. Determine the **Interquartile Range (IQR)**.

4. Identify if there are any **outliers** (using the $1.5 \times IQR$ rule).
 5. What percentage of the data falls between Q1 and Q3?
2. **Box-and-Whisker Plot with Outliers:** A network administrator records the data transmission times (in milliseconds) for various network packets. The times are:

25, 28, 30, 33, 35, 37, 40, 42, 45, 50, 55, 60, 62, 120

1. Find the **five-number summary** (Minimum, Q1, Median, Q3, Maximum).
2. Compute the **Interquartile Range (IQR)**.
3. Identify any **outliers** using the $1.5 \times IQR$ rule.
4. Construct a **Box-and-Whisker Plot** for the given data.

Conceptual Questions

1. Why is the **Interquartile Range (IQR)** useful for detecting outliers?
2. How do you interpret a Box-and-Whisker Plot that has a longer upper whisker than a lower whisker?
3. If a new data point, 130 ms, is added, how does it affect the Box-and-Whisker Plot?